

## Year 7 · Chapter 11 · Probability: answers (teacher copy)

### 11.01 Sample space (booklet p. 3)

#### WE DO examples (teacher)

- Column 1, E1: tossing a coin → 2
- Column 1, E2: rolling a die → 6
- Column 2, E1: rolling a die → { 1, 2, 3, 4, 5, 6 }; 6 outcomes
- Column 2, E2: choosing a letter from CAT → { C, A, T }; 3 outcomes
- Column 3, E1: toss a coin and roll a die →  $2 \times 6 = 12$ ; e.g. H1, H2, ..., T6
- Column 3, E2: toss two coins →  $2 \times 2 = 4$ ; e.g. HH, HT, TH, TT

#### Questions (in order: Easy, Medium, Challenging)

- |       |                   |                        |
|-------|-------------------|------------------------|
| 1. 7  | 11. 8             | 18. 2, 3, 5, 7, 11     |
| 2. 12 | 12. 5             | 19. B, O, K            |
| 3. 52 | 13. 7             | 20. 5, 10, 15, 20, 25, |
| 4. 26 | 14. hearts,       | 30                     |
| 5. 5  | diamonds,         | 21. 1, 2, 3, 4, 6, 12  |
| 6. 4  | clubs, spades     | 22. 36                 |
| 7. 5  | 15. 1, 3, 5, 7, 9 | 23. 12                 |
| 8. 10 | 16. Saturday,     | 24. 8                  |
| 9. 5  | Sunday            | 25. 14                 |
| 10. 4 | 17. a, e, i, o, u | 26. 10                 |

### 11.02 Probability (booklet p. 4)

#### WE DO examples (teacher)

- Column 1, E1:  $P(a \ 4) \rightarrow \frac{1}{6}$
- Column 1, E2:  $P(\text{an even number}) \rightarrow \frac{1}{2}$
- Column 2, E1: A bag has 3 green and 9 red counters.  
 $P(\text{green}) \rightarrow 3 \text{ green out of } 12; P = \frac{3}{12} = \frac{1}{4}$
- Column 2, E2: A bag has 1 green and 5 yellow counters.  
 $P(\text{green}) \rightarrow 1 \text{ green out of } 6; P = \frac{1}{6} = \frac{1}{6}$
- Column 3, E1:  $P(\text{green})$ , as a fraction and a percentage → 2 of the 6 sectors are green;  $P = \frac{1}{3} = 33.3\%$
- Column 3, E2:  $P(\text{yellow})$ , as a fraction and a percentage → 2 of the 8 sectors are yellow;  $P = \frac{1}{4} = 25\%$

#### Questions (in order: Easy, Medium, Challenging)

- |                  |                   |                            |
|------------------|-------------------|----------------------------|
| 1. $\frac{1}{2}$ | 8. $\frac{1}{3}$  | 15. $\frac{8}{9}$          |
| 2. $\frac{1}{3}$ | 9. $\frac{2}{3}$  | 16. $\frac{4}{5}$          |
| 3. $\frac{1}{3}$ | 10. $\frac{1}{3}$ | 17. $\frac{1}{4}$          |
| 4. 0             | 11. 1             | 18. $\frac{7}{12}$         |
| 5. $\frac{1}{3}$ | 12. $\frac{1}{3}$ | 19. $\frac{1}{2}$          |
| 6. $\frac{1}{2}$ | 13. $\frac{3}{7}$ | 20. $\frac{1}{3} = 33.3\%$ |
| 7. $\frac{5}{6}$ | 14. $\frac{3}{7}$ | 21. $\frac{1}{4} = 25\%$   |

### 11.03 The probability scale (booklet p. 5)

#### WE DO examples (teacher)

- Column 1, E1: 0.75 → likely
- Column 1, E2: 5% → unlikely
- Column 2, E1: 2 red and 5 blue counters:  $P(\text{red}) \rightarrow P(\text{red}) = \frac{2}{7} = \frac{2}{7}$ ; unlikely
- Column 2, E2: 0 red and 6 blue counters:  $P(\text{red}) \rightarrow P(\text{red}) = \frac{0}{6}$ ; impossible

5. Column 3, E1:  $\frac{2}{5}$  or 55%? →  $\frac{2}{5} = 0.4$ , 55% = 0.55; more likely: 55%

6. Column 3, E2:  $\frac{3}{8}$  or 45%? →  $\frac{3}{8} = 0.375$ , 45% = 0.45; more likely: 45%

#### Questions (in order: Easy, Medium, Challenging)

- |                |                              |                              |
|----------------|------------------------------|------------------------------|
| 1. likely      | 17. unlikely                 | 32. $\frac{2}{5}$ , unlikely |
| 2. unlikely    | 18. likely                   | 33. 1, certain               |
| 3. unlikely    | 19. likely                   | 34. $\frac{1}{5}$ , unlikely |
| 4. unlikely    | 20. likely                   | 35. 0, impossible            |
| 5. unlikely    | 21. likely                   | 36. $\frac{3}{8}$            |
| 6. certain     | 22. unlikely                 | 37. 70%                      |
| 7. even chance | 23. even chance              | 38. $\frac{7}{10}$           |
| 8. unlikely    | 24. likely                   | 39. $\frac{7}{10}$           |
| 9. likely      | 25. certain                  | 40. 65%                      |
| 10. unlikely   | 26. impossible               | 41. 35%                      |
| 11. likely     | 27. unlikely                 | 42. $\frac{2}{3}$            |
| 12. unlikely   | 28. likely                   | 43. $\frac{2}{5}$            |
| 13. unlikely   | 29. $\frac{4}{9}$ , unlikely |                              |
| 14. likely     | 30. $\frac{3}{4}$ , likely   |                              |
| 15. impossible | 31. 0, impossible            |                              |
| 16. likely     |                              |                              |

### 11.04 Experimental probability (booklet p. 6)

#### WE DO examples (teacher)

- Column 1, E1: heads in 36 trials → 18
- Column 1, E2: a 1 or 2 on a die in 48 trials → 16
- Column 2, E1: 25 heads in 50 tosses →  $\frac{25}{50} = \frac{1}{2}$
- Column 2, E2: 3 heads in 50 tosses →  $\frac{3}{50} = \frac{3}{50}$
- Column 3, E1: A 6 came up 25 times in 120 rolls. Is the die fair? → experiment:  $25 \div 120 = 20.8\%$ ; theory:  $1 \div 6 = 16.7\%$ ; close, so probably fair
- Column 3, E2: A 6 came up 9 times in 30 rolls. Is the die fair? → experiment:  $9 \div 30 = 30\%$ ; theory:  $1 \div 6 = 16.7\%$ ; not close: roll more times to check

#### Questions (in order: Easy, Medium, Challenging)

- |        |                     |                      |
|--------|---------------------|----------------------|
| 1. 28  | 14. 15              | 25. $\frac{1}{10}$   |
| 2. 6   | 15. 25              | 26. $\frac{43}{50}$  |
| 3. 10  | 16. 9               | 27. $\frac{19}{20}$  |
| 4. 27  | 17. 13              | 28. $\frac{31}{100}$ |
| 5. 27  | 18. 18              | 29. $\frac{23}{40}$  |
| 6. 4   | 19. 28              | 30. 26.7% vs 16.7%   |
| 7. 3   | 20. 6               | 31. 16.7% vs 16.7%   |
| 8. 8   | 21. 22              | 32. 15% vs 16.7%     |
| 9. 16  | 22. $\frac{19}{50}$ | 33. 20% vs 16.7%     |
| 10. 9  | 23. $\frac{19}{60}$ | 34. 16.7% vs 16.7%   |
| 11. 25 | 24. $\frac{17}{50}$ | 35. 23.3% vs 16.7%   |
| 12. 19 |                     |                      |
| 13. 3  |                     |                      |

### 11.05 Complementary events (booklet p. 7)

#### WE DO examples (teacher)

- Column 1, E1:  $P(E) = 0.38 \rightarrow 0.63$
- Column 1, E2:  $P(E) = 0.2 \rightarrow 0.8$
- Column 2, E1: rolling a 6 → complement: not rolling a 6;  $P = 1 - \frac{1}{6} = \frac{5}{6}$
- Column 2, E2: tossing heads → complement: tossing tails;  $P = 1 - \frac{1}{2} = \frac{1}{2}$

5. Column 3, E1: 9 red, 9 blue, 9 green: P(not green) → total:  
27; not green:  $27 - 9 = 18$ ;  $P = \frac{18}{27} = \frac{2}{3}$
6. Column 3, E2: 3 red, 1 blue, 7 green: P(not blue) → total:  
11; not blue:  $11 - 1 = 10$ ;  $P = \frac{10}{11}$

**Questions (in order: Easy, Medium, Challenging)**

- |                  |                    |                   |
|------------------|--------------------|-------------------|
| 1. 0.33          | 6. $\frac{1}{2}$   | 11. $\frac{7}{8}$ |
| 2. 0.25          | 7. $\frac{5}{6}$   | 12. 0.14          |
| 3. $\frac{1}{3}$ | 8. 0.89            | 13. 0.75          |
| 4. $\frac{7}{9}$ | 9. $\frac{3}{5}$   | 14. 0.78          |
| 5. 0.08          | 10. $\frac{1}{12}$ | 15. $\frac{3}{4}$ |

16.  $\frac{1}{6}$
17. 0.2
18.  $\frac{5}{8}$
19. 0.71
20. 0.86
21.  $\frac{7}{10}$
22.  $\frac{6}{7}$
23.  $\frac{9}{10}$
24.  $\frac{4}{5}$
25.  $\frac{1}{9}$

26.  $\frac{2}{7}$
27.  $\frac{5}{9}$
28. 0.13
29. 0.5
30.  $\frac{1}{10}$
31. a black card,  $\frac{1}{2}$
32. an odd number,  
 $\frac{1}{2}$
33. a club,  
diamond or  
spade,  $\frac{3}{4}$

34. rolling 3 or  
more,  $\frac{2}{3}$
35. a consonant,  $\frac{4}{5}$
36.  $\frac{1}{2}$
37.  $\frac{5}{7}$
38.  $\frac{11}{13}$
39.  $\frac{4}{7}$
40.  $\frac{3}{5}$